

n	Numeric literals	
$Location, l$	Store locations	
b	$::=$ $ $ true $ $ false	booleans
$operations, op$	$::=$ $ $ $+$ $ $ $\geq$	operations
$expr, e$	$::=$ $ $ n $ $ b $ $ $e_1 \ op \ e_2$ $ $ if e_1 then e_2 else e_3 $ $ $l := e_1$ $ $ $!l$ $ $ skip $ $ $e_1; e_2$ $ $ while e_1 do e_2 $ $ (e)	expressions
v	$::=$ $ $ n $ $ b $ $ skip	values
$store, s$	$::=$ $ $ empty $ $ $s, l \mapsto n$	stores
$config, c$	$::=$ $ $ $\langle e, s \rangle$	configurations
typ, T	$::=$ $ $ int $ $ bool $ $ unit	types
$Tloc$	$::=$ $ $ intref	types of locations
$type_assumption, ta$	$::=$ $ $ $l : Tloc$	type assumption
Γ	$::=$ $ $ empty $ $ Γ, ta	type environments
$terminals$	$::=$ $ $ $+$	

		$\geq$	
		$\vdash$	
		$\mapsto$	
		$;$	
		$\longrightarrow$	
		$\langle$	
		$\rangle$	
<i>formula</i>	$::=$		
		<i>judgement</i>	
		$n_1 + n_2 = n_3$	
		$n_1 \geq n_2 = b$	
		$s(l) = n$	
		$\Gamma(l) = Tloc$	
<i>Jop</i>	$::=$		
		$c \longrightarrow c'$	c reduces to c'
		$\Gamma \vdash e : T$	
<i>judgement</i>	$::=$		
		<i>Jop</i>	
<i>user_syntax</i>	$::=$		
		n	
		<i>Location</i>	
		b	
		<i>operations</i>	
		<i>expr</i>	
		v	
		<i>store</i>	
		<i>config</i>	
		<i>typ</i>	
		$Tloc$	
		<i>type_assumption</i>	
		Γ	
		<i>terminals</i>	
		<i>formula</i>	

$\boxed{c \longrightarrow c'}$ c reduces to c'

$$\begin{array}{c}
\frac{n_1 + n_2 = n}{\langle n_1 + n_2, s \rangle \longrightarrow \langle n, s \rangle} \quad \text{OP_PLUS} \\
\frac{n_1 \geq n_2 = b}{\langle n_1 \geq n_2, s \rangle \longrightarrow \langle b, s \rangle} \quad \text{OP_GTEQ} \\
\frac{\langle e_1, s \rangle \longrightarrow \langle e'_1, s' \rangle}{\langle e_1 \text{ op } e_2, s \rangle \longrightarrow \langle e'_1 \text{ op } e_2, s' \rangle} \quad \text{OP1} \\
\frac{\langle e_2, s \rangle \longrightarrow \langle e'_2, s' \rangle}{\langle e_1 \text{ op } e_2, s \rangle \longrightarrow \langle e_1 \text{ op } e'_2, s' \rangle} \quad \text{OP2} \\
\frac{s(l) = n}{\langle !l, s \rangle \longrightarrow \langle n, s \rangle} \quad \text{DEREF}
\end{array}$$

$$\begin{array}{c}
\frac{}{\langle l := n, s \rangle \longrightarrow \langle \mathbf{skip}, s, l \mapsto n \rangle} \text{ ASSIGN1} \\
\frac{\langle e, s \rangle \longrightarrow \langle e', s' \rangle}{\langle l := e, s \rangle \longrightarrow \langle l := e', s' \rangle} \text{ ASSIGN2} \\
\frac{}{\langle \mathbf{skip}; e, s \rangle \longrightarrow \langle e, s \rangle} \text{ SEQ1} \\
\frac{\langle e_1, s \rangle \longrightarrow \langle e'_1, s' \rangle}{\langle e_1; e_2, s \rangle \longrightarrow \langle e'_1; e_2, s' \rangle} \text{ SEQ2} \\
\frac{}{\langle \mathbf{if true then } e_1 \mathbf{ else } e_2, s \rangle \longrightarrow \langle e_1, s \rangle} \text{ IF1} \\
\frac{}{\langle \mathbf{if false then } e_1 \mathbf{ else } e_2, s \rangle \longrightarrow \langle e_2, s \rangle} \text{ IF2} \\
\frac{\langle e, s \rangle \longrightarrow \langle e', s' \rangle}{\langle \mathbf{if } e \mathbf{ then } e_1 \mathbf{ else } e_2, s \rangle \longrightarrow \langle \mathbf{if } e' \mathbf{ then } e_1 \mathbf{ else } e_2, s' \rangle} \text{ IF3} \\
\frac{}{\langle \mathbf{while } e_1 \mathbf{ do } e_2, s \rangle \longrightarrow \langle \mathbf{if } e_1 \mathbf{ then } (e_2; \mathbf{while } e_1 \mathbf{ do } e_2) \mathbf{ else skip}, s \rangle} \text{ WHILE}
\end{array}$$

$$\boxed{\Gamma \vdash e : T}$$

$$\begin{array}{c}
\frac{}{\Gamma \vdash n : \mathbf{int}} \text{ TY_INT} \\
\frac{}{\Gamma \vdash b : \mathbf{bool}} \text{ TY_BOOL} \\
\frac{\Gamma \vdash e_1 : \mathbf{int} \quad \Gamma \vdash e_2 : \mathbf{int}}{\Gamma \vdash e_1 + e_2 : \mathbf{int}} \text{ TY_OP_PLUS} \\
\frac{\Gamma \vdash e_1 : \mathbf{int} \quad \Gamma \vdash e_2 : \mathbf{int}}{\Gamma \vdash e_1 \geq e_2 : \mathbf{bool}} \text{ TY_OP_GTEQ} \\
\frac{\Gamma \vdash e_1 : \mathbf{bool} \quad \Gamma \vdash e_2 : T \quad \Gamma \vdash e_3 : T}{\Gamma \vdash \mathbf{if } e_1 \mathbf{ then } e_2 \mathbf{ else } e_3 : T} \text{ TY_IF} \\
\frac{\Gamma(l) = \mathbf{intref} \quad \Gamma \vdash e : \mathbf{int}}{\Gamma \vdash l := e : \mathbf{unit}} \text{ TY_ASSIGN} \\
\frac{\Gamma(l) = \mathbf{intref}}{\Gamma \vdash !l : \mathbf{int}} \text{ TY_DEREF} \\
\frac{}{\Gamma \vdash \mathbf{skip} : \mathbf{unit}} \text{ TY_SKIP} \\
\frac{\Gamma \vdash e_1 : \mathbf{unit} \quad \Gamma \vdash e_2 : T}{\Gamma \vdash e_1; e_2 : T} \text{ TY_SEQ} \\
\frac{\Gamma \vdash e_1 : \mathbf{bool} \quad \Gamma \vdash e_2 : \mathbf{unit}}{\Gamma \vdash \mathbf{while } e_1 \mathbf{ do } e_2 : \mathbf{unit}} \text{ TY_WHILE}
\end{array}$$

Definition rules: 23 good 0 bad
 Definition rule clauses: 45 good 0 bad